Construction of a flavonoid database for assessing intake in a population-based sample of women on Long Island, New York.

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Flavonoids have been hypothesized to reduce cancer risk. Previous epidemiological studies conducted to evaluate this hypothesis have not assessed all flavonoids, including classes that could contribute to intake among Americans, which would result in an underestimation of intake. This misclassification could mask variability among individuals, resulting in attenuated effect estimates for the association between flavonoids and cancer. To augment flavonoid and lignan intake estimates, we developed a database that can be used in conjunction with a food-frequency questionnaire (FFQ). Coupling information derived from the available literature with the U.S. Department of Agriculture databases, we estimated content of 6 flavonoid classes and lignans for 50 food group items. We combined these estimates with responses from a modified Block FFQ that was self-completed in 1996-1997 by a population-based sample of women without breast cancer on Long Island, New York (n = 1,500). Total flavonoid and lignan content of food items ranged from 0 to 129 mg/100 g, and the richest sources were tea, cherries, and grapefruit. Individual intake estimates, from highest to lowest, were flavan-3-ols, flavanones, flavonols, lignans, isoflavones, anthocyanidins, and flavones. Each class of flavonoids and lignans exhibited a wide range of intake levels. This database is useful to quantify flavonoid and lignan intake for other observational studies conducted in the United States that utilize the Block FFQ.